

Technical Document #108: Natural Language Processing - Comprehensive Overview

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The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys. It is the core component of transformer models.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.
- Text summarization generates a concise summary of a longer text.
- Word embeddings represent words as dense vectors in continuous space.